

SHORT COMMUNICATION

Topical Aloe Vera (*Aloe barbadensis* Miller) Extract Does Not Accelerate the Oral Wound Healing in Rats

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The effect of topical application of Aloe Vera (*Aloe barbadensis* Miller) extract was assessed on the healing of rat oral wounds in an *in vivo* model using 72 male Wistar rats divided into three groups ($n = 24$): control, placebo and Aloe Vera (0.5% extract hydroalcoholic). Traumatic ulcers were caused in the dorsum of the tongue using a 3-mm punch tool. The Aloe Vera and placebo group received two daily applications. The animals were sacrificed after 1, 5, 10 and 14 days. Clinical analysis (ulcer area and percentage of repair) and histopathological analysis (degree of re-epithelialization and inflammation) were performed. The comparison of the differences between scores based on group and experimental period, both in quantitative and semi-quantitative analyses, was performed using the Kruskal–Wallis test. The significance level was 5%. On day 1, all groups showed predominantly acute inflammatory infiltrate. On day 5, there was partial epithelialization and chronic inflammatory infiltrate. On the days 10 and 14 total repair of ulcers was observed. There was no significant difference between groups in the repair of mouth ulcers. It is concluded that treatment using Aloe Vera as an herbal formulation did not accelerate oral wound healing in rats. Copyright © 2015 John Wiley & Sons, Ltd.

Keywords: wound healing; oral ulcer; aloe vera; rat; phytotherapy.

INTRODUCTION

Oral ulcerative lesions are common in dental practice; however, their diagnoses are sometimes made difficult because of the large number of diseases that lead to this condition. Usually, they present traumatic or autoimmune etiology (Chavan *et al.*, 2012; Felix *et al.*, 2012; Leão *et al.*, 2007). The treatment of these mouth ulcers has not been fully established, and several protocols using topical or systemic drugs have been described (Chavan *et al.*, 2012; Felix *et al.*, 2012; Leão *et al.*, 2007). Frequently, the aim is to relieve symptoms and accelerate the repair of ulceration to allow tissue reconstitution (Mandelbaum *et al.*, 2003; Eming *et al.*, 2007; de Carvalho *et al.*, 2015). The use of herbal medicines in wound healing has been extensively studied, because plants have several active ingredients such as alkaloids, triterpenes and biomolecules that affect one or more repair phases (Fernandes *et al.*, 2010; Martins *et al.*, 2009, 2011; Mogosan *et al.*, 2013; O'Hara *et al.*, 1998).

Among these herbal remedies, Aloe Vera is a plant that has been used in traditional medicine by many cultures in an attempt to cure numerous diseases (Chithra *et al.*, 1998). Also called *Aloe barbadensis* Miller, it belongs to the family Liliaceae, which is approximately 360 species. Aloe Vera is a plant that grows easily in hot and dry climates (Chithra *et al.*, 1998). This plant has several important properties and may interfere in the process of tissue repair. The action of Aloe Vera in tissue repair has been attributed to the fact that this plant stimulates macrophage function and nitric oxide production, as apart from increasing collagen production in the granulation tissue. Moreover, this plant species has vasodilator, antibacterial and antifungal action (Herlihy *et al.*, 1998; Vogler and Ernest, 1999). This property is cited in the literature in both topical and systemic applications; however, controversial results have been described. Some studies that investigated the topical use of Aloe Vera showed acceleration of wound healing (Choonhakarn *et al.*, 2007; Eshghi *et al.*, 2010; Heggers *et al.*, 1997; Khorasani *et al.*, 2009; Oryan *et al.*, 2014; Rahmani *et al.*, 2014), but in other studies wound healing was delayed (Schmidt and Greenspoon, 1991; Paulsen *et al.*, 2005). The aim of this study was to evaluate the clinical and histopathological aspects of topical treatment with 0.5% Aloe Vera hydroalcoholic extract in oral wound healing in rats.

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MATERIALS AND METHODS

Preparation of 0.5% extract hydroalcoholic (*Aloe barbadensis* Miller). Commercially available concentrated Aloe Vera granulated powder of certified quality was used (Viafarma LTDA, freeze dried powder, 200:1 concentration). The oral topical cream was always prepared using powder from the same batch, in the same compounding pharmacy and under strict quality control. The components of the cream are: hydroalcoholic extract (solvent), nipagin (preservative), nipazol (preservative), phenonip (preservative), propylene glycol (vehicle), glycerol (vehicle) and distilled water. The final concentration of Aloe Vera was 0.5% as recommended for topical use (Eshghi *et al.*, 2010; Khorasani *et al.*, 2009; Syed *et al.*, 1996). The placebo was prepared using the same components of the gel, without Aloe Vera extract.

Animals. All experiments were carried out in accordance with the Guide for the Care and Use of Laboratory Animals and received approval from the ethics committee of the Porto Alegre University Hospital (HCPA, Brazil) (12–0134). Seventy-two male rats (*Rattus norvegicus albinus*, *rodentia mammalia*—wistar lineage) weighing 150 to 200 g were kept under standard conditions of temperature (20 to 24 °C) and light/dark cycle, with solid chow and water *ad libitum*. The animals were randomly divided into three groups of 24 animals each: Control group (no treatment, only daily handling), Placebo Group (topical treatment with a hydroalcoholic extract without Aloe Vera) and Aloe Vera Group (topical treatment with 0.5% Aloe Vera extract hydroalcoholic).

Wound procedure and treatment protocol. Under aseptic conditions, the three groups were anesthetized with an intraperitoneal administration of ketamine (0.1 ml/100 g) and xylazine (0.05 ml/100 g). Traumatic ulcers

measuring 3 mm in diameter were inflicted on the dorsum of the tongue following a standard punch biopsy technique. The animals in the Placebo and Aloe Vera groups received two daily applications of the product using a swab so as fully cover the wound at 12-h intervals throughout the experimental period. During application animals were kept under isoflurane inhalant anesthesia for 14 consecutive days. Six rats in each group were euthanized using a CO₂ chamber on days 1, 5, 10 and 14 after the surgical procedure. After, the wound area was measured and photographed.

Clinical analysis. Ulcer area was calculated based on length and width using a digital caliper. These measurements were multiplied by the area in square centimeters. The percentages of wound healing and healing time were recorded as described elsewhere (Khorasani *et al.*, 2009).

Histopathological analysis. After euthanasia, the tongues were removed and fixed in a 10% buffered formalin solution for 48 h. After washing with water, the specimens were dehydrated and embedded in paraffin. Slices measuring 5 µm in thickness were obtained and stained with hematoxylin–eosin. Descriptive analysis of each group/evaluation time was performed, followed by a semi-quantitative analysis. The analysis was done by two pathologists to reach a consensual final score. The degree of reepithelialization was determined a grading system (0 to 4), as described by elsewhere (Sinha and Gallagher, 2003). Inflammation was also evaluated according to a grading system, as described by Kumar *et al.* (2003) and Camacho-Alonso and Lopez-Jornet (2007).

Statistical analysis. The data were expressed as mean and standard deviation values. The SPSS version 18.0 was employed for the statistical analysis. Groups,

Table 1. Clinical evaluation of mean and standard error of wound area (in mm²)

Group	Day 1	Day 5	Day 10	Day 14
Control	5.2 ± 1.19 aA	3.18 ± 0.69 aA	0.0 ± 0.0 bA	0.0 ± 0.0 bA
Placebo	2.89 ± 0.96 aA	1.31 ± 1.31 aA	0.0 ± 0.0 aA	0.0 ± 0.0 aA
Aloe Vera	3.02 ± 0.8 aA	1.01 ± 0.81 aA	0.02 ± 0.03 aA	0.0 ± 0.0 aA

Different lowercase letters on lines (intra-group analysis) denote significant difference ($p < 0.05$, Kruskal–Wallis test); different uppercase letters in columns (inter-group analysis) denote significant difference ($p < 0.05$, Kruskal–Wallis test).

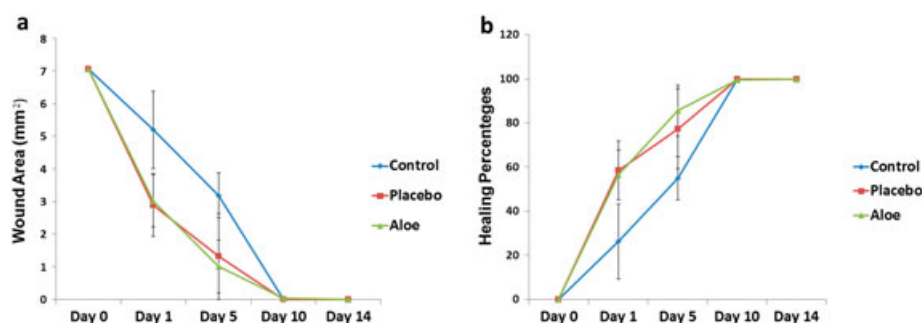


Figure 1. (a) Clinical evaluation of mean and standard error of ulcer area (in mm²) in all groups; (b) clinical evaluation of percentage of wound healing (%) in all groups during the experimental time.

Table 2. Clinical evaluation of mean and standard error of healing percentage

Group	Day 1	Day 5	Day 10	Day 14
Control	26.2 ± 16.94 aA	54.9 ± 9.82 aA	100 ± 0.0 bA	100 ± 0.0 bA
Placebo	58.5 ± 13.3 aA	77.4 ± 18.06 aA	100 ± 0.0 aA	100 ± 0.0 aA
Aloe Vera	56.5 ± 11.3 aA	85.5 ± 11.5 aA	99.6 ± 0.51 aA	100 ± 0.0 aA

Different lowercase letters on lines (intra-group analysis) denote significant difference ($p < 0.05$, Kruskal–Wallis test); different uppercase letters in columns (inter-group analysis) denote significant difference ($p < 0.05$, Kruskal–Wallis test).

Table 3. Histopathological evaluation of degree of reepithelialization (mean and standard error)

Group	Day 1	Day 5	Day 10	Day 14
Control	0.75 ± 0.95 aA	2.75 ± 0.5 aA	3.5 ± 0.5 aA	3.5 ± 0.5 aA
Placebo	1.0 ± 0.81 aA	2.0 ± 0.81 aA	3.75 ± 0.5 aA	4.0 ± 0.0 aA
Aloe Vera	1.25 ± 0.95 aA	1.75 ± 0.5 aA	3.25 ± 0.5 aA	3.75 ± 0.5 aA

Different lowercase letters on lines (intra-group analysis) denote significant difference ($p < 0.05$, Kruskal–Wallis test); different uppercase letters in columns (inter-group analysis) denote significant difference ($p < 0.05$, Kruskal–Wallis test).

Table 4. Histopathological evaluation of degree of inflammation (mean and standard error)

Group	Day 1	Day 5	Day 10	Day 14
Control	1.0 ± 0.0 aA	2.5 ± 0.5 aA	3.75 ± 0.5 aA	3.75 ± 0.5 aA
Placebo	1.25 ± 0.5 aA	1.75 ± 0.95 aA	4.0 ± 0.0 aA	4.0 ± 0.0 aA
Aloe Vera	1.5 ± 0.5 aA	2.5 ± 1.0 aA	3.25 ± 0.5 aA	3.5 ± 0.5 aA

Different lowercase letters on lines (intra-group analysis) denote significant difference ($p < 0.05$, Kruskal–Wallis test); different uppercase letters in columns (inter-group analysis) denote significant difference ($p < 0.05$, Kruskal–Wallis test).

evaluation times and the interaction between group and evaluation time were compared using the Kruskal–Wallis test. The significance level was set at 5% ($p < 0.05$).

RESULTS AND DISCUSSION

Table 1 and Fig. 1a illustrate the clinical analysis of wound area. There were no differences between groups and no detectable signs of repair on Day 1. A decrease in mean wound area was found in all groups on Day 5, with the smallest area in the Aloe Vera group; however, no statistical difference was observed. On Day 10 and Day 14 all groups exhibited repaired lesions with no statistically significant differences between groups.

The percentages of wound healing analyzed on Day 1 revealed no significant differences between groups. An increase in percentage of wound healing was gradually observed on Days 5, 10 and 14. On Day 10 a total closure of wound was observed in all groups, without differences (Table 2 and Fig. 1b).

Histopathological analysis

Regarding histopathological analysis, on day 1 all groups showed predominantly acute inflammatory infiltrate. On day 5, there was partial epithelialization and chronic inflammatory infiltrate. On the days 10 and 14 all groups exhibited reepithelialization covering the entire wound.

Table 3 demonstrates the mean results of the grading system analysis of reepithelialization, with no statistically significant differences between groups in all periods. Table 4 demonstrates the mean values of the grading system analysis of inflammation, no statistically significant differences between groups, across all periods.

In our study, the Aloe Vera extract tested revealed a moderate inflammation process in the initial days of ulcer healing followed by the reepithelialization, formation of granulation tissue and remodeling of the connective tissue. However, these effects were similar in the control and placebo groups. Although the 0.5% Aloe Vera extract hydroalcoholic did not promote an increase in wound healing, it did not present any side effect. One possible explanation for the non-healing effect of Aloe Vera tested in our study could be associated with product manufacture methods and concentration of the phytotherapeutic agent, when compared to the literature. One of the most discussed components of Aloe Vera that could lead to a different response is polysaccharides (Paulsen *et al.*, 2005). Whatever the reason, commercial Aloe Vera products have been shown to contain widely varying levels of polysaccharides that have been attributed to differing climatic conditions, different gel preparation techniques and product storage. A low polysaccharide content would make the Aloe Vera gel more similar to the placebo gel. Unfortunately, our Aloe Vera gel was not tested for its content.

Other important point that could influence tissue response is Aloe Vera concentration. We decided to use the 0.5% Aloe Vera concentration based on some studies that previously indicate positive results (Eshghi

et al., 2010; Khorasani *et al.*, 2009; Syed *et al.*, 1996) and because local side effects like erythema and desquamation were evident when the highest concentration (98%) was used. However, based on our results, we believe that 0.5% Aloe Vera for oral wound healing was not sufficient to induce faster healing. This could be explained based on the fact that the healing of oral mucosa is faster than in the skin, and that perhaps low concentrations of Aloe Vera are not sufficient to stimulate the oral epithelium, which usually has higher proliferative index.

CONCLUSION

In conclusion, the findings of the present study show that the 0.5% Aloe Vera hydroalcoholic extract does not

promote any local side effect, but it is not useful to improve the healing of oral ulcers. Other concentrations of Aloe Vera or different vehicles need to be investigated.

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Conflict of Interest

The authors have declared that there is no conflict of interest.

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